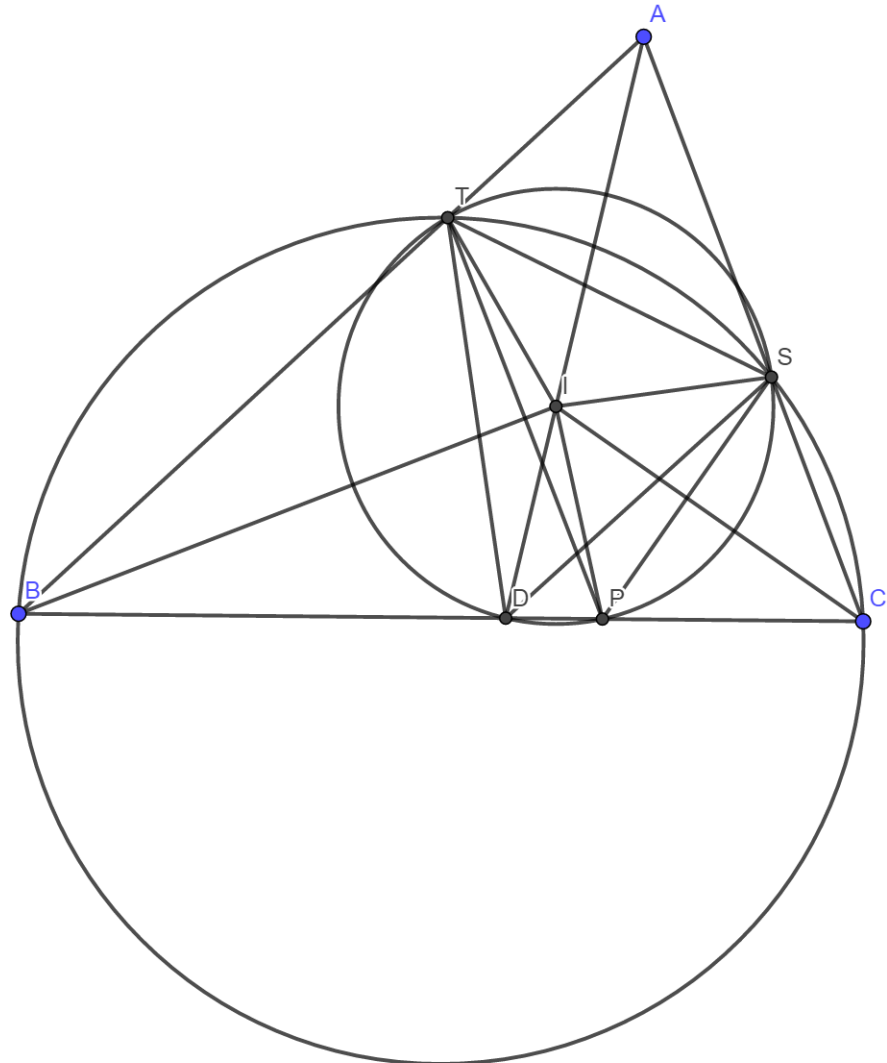


4.3.13



Obviously, S and T are reflections of P in CI and BI . Then $\angle ASI = 180^\circ - \angle CSI = 180^\circ - \angle CPI = \angle BPI = \angle BTI$, so A, S, I, T are concyclic. Then $IS = IT = IP$. Then $\angle AIT = \angle AST = \angle ABC$, so B, D, I, T are concyclic. So $ID = IT$, D, P, S, T are concyclic with centre I .